

Estimating mixing properties of local Hamiltonian dynamics
and continuous quantum random walks is PSPACE-hard

Measuring local observables could probabilistically solve
PSPACE

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Observable, Hamiltonian, spectral resolution

finite dimensional Hilbert space $\mathcal{H}$

observable / **Hamiltonian** : Hermitian operator A on $\mathcal{H}$

continuous quantum random walk : A is an adjacency matrix

spectral resolution of A :

different eigenvalues of A : (λ)

orthogonal projections onto the corresponding eigenspaces of A : (P_λ)

Mixing properties of Hamiltonian dynamics and continuous quantum random walks

Time-average of a state ρ on $\mathcal{H}$:

$$\bar{\rho} := \lim_{T \rightarrow \infty} \frac{1}{T} \int_{t=0}^T e^{-iAt} \rho e^{iAt} dt$$

von-Neumann entropy $S(\bar{\rho})$ of the time-average

$$S(\bar{\rho}) := \text{tr}(\bar{\rho} \log_2 \bar{\rho})$$

motivation for considering $S(\bar{\rho})$:

quantifies “how much and how uniformly” the orbit $\rho_t := e^{-iAt} \rho e^{iAt}$ fills out the state space

quantifies “how far” a pure state $\rho := |\Phi\rangle\langle\Psi|$ is from all eigenvectors

Measurement of observables

Postulate: Measurement of A with accuracy Δ :

for all states ρ on $\mathcal{H}$ the probability to obtain an outcome in the interval

$$[\lambda - \Delta, \lambda + \Delta]$$

is at least

$$(3/4) \operatorname{tr}(\rho P_\lambda)$$

Local operators

$\mathcal{H}$ tensor product space of n Hilbert spaces

$$\mathcal{H} := \mathcal{H}_1 \otimes \mathcal{H}_2 \otimes \cdots \otimes \mathcal{H}_n$$

A is k -local:

A sum of operators acting on at most k tensor components non-trivially

example for 2-local Hamiltonians:

Ising interactions, coupling topology specified by a graph $G = (V, E)$

$$\sum_{(k,l) \in E} \sigma_z^{(k)} \sigma_z^{(l)}$$

$$\sigma_z^{(k)} = \mathbf{1} \otimes \cdots \otimes \mathbf{1} \otimes \underbrace{\sigma_z}_{k\text{th qubit}} \otimes \mathbf{1} \otimes \cdots \otimes \mathbf{1}$$

we want to prove

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Measuring local observables could probabilistically solve
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Outline of the proof:

recall definition of PSPACE (based on irreversible Turing machines)

transform irreversible TMs to reversible TM

construct quantum circuit that simulate the computational steps of the
reversible TMs

encode the sequence of gates of these quantum circuits in local
operators

→ desired observables, Hamiltonians, adjacency matrices

Definition of PSPACE

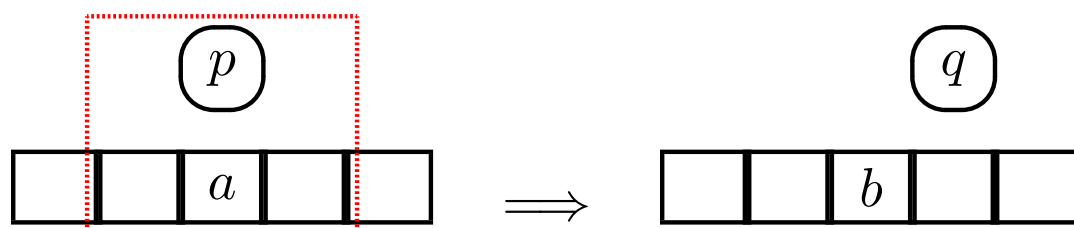
class of all languages recognizable by polynomial space bounded DTMs

Deterministic Turing Machine:

head (internal states Q)

tape divided into cells (contain symbols of the alphabet Σ)

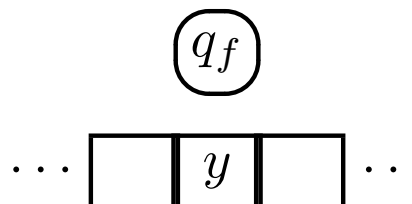
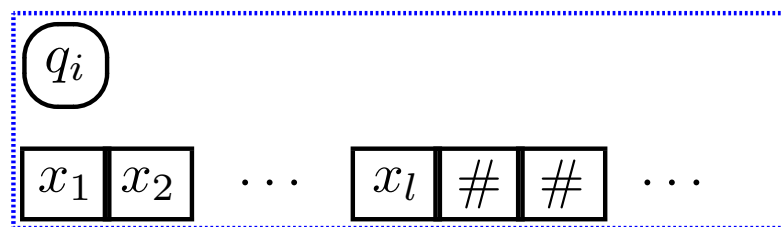
local transition function $Q \times \Sigma \rightarrow Q \times \Sigma \times \{-1, 0, +1\}$



input: $x_1 x_2 \dots x_l$

$\text{poly}(l)$

output: symbol in current cell when
RTM goes into a final state q_f



Quantified Boolean Formula Problem

is a **PSPACE-complete** problem

quantified Boolean formula:

$$Q_1 v_1 Q_2 v_2 \cdots Q_n v_n B(v_1, v_2, \dots, v_n)$$

Q_i : quantifiers (either $\forall$ or $\exists$)

$B(v_1, v_2, \dots, v_n)$: Boolean formula (without quantifiers)

in the variables $v_1, v_2, \dots, v_n$

quantified Boolean formula problem:

determine if a quantified formula is true

it can be solved within space $O(l^2)$ (l input length) by an irreversible TM

Reversible Turing machines

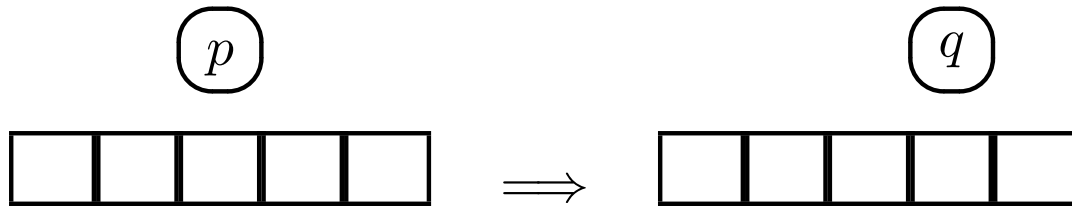
Lange et al., “Reversible space equals deterministic space”, J. Comp. & Syst. Sci., 2000.

irreversible TMs can be simulated space-efficiently by reversible TMs

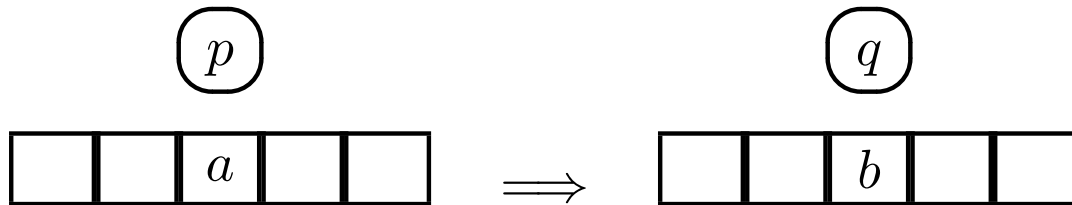
explicit construction of the RTM

each transition is

either **moving** $p \rightarrow (q, \pm 1)$



or **read-and-write** $(p, a) \rightarrow (q, b)$



Quantum circuit simulating the RTM (first idea)

registers: `head`, `tape_index`, `ACC`, `cell[i]` $i = 0, \dots, N$

U_{moving} implements $p \rightarrow (q, \pm 1)$

$$|p\rangle_{\text{head}} \otimes |i\rangle_{\text{tape_index}} \rightarrow |q\rangle_{\text{head}} \otimes |i \pm 1\rangle_{\text{tape_index}}$$

$U_{\text{r\&w}}$ implements $(p, a) \rightarrow (q, b)$

- swap `ACC` and `cell[i]` (current cell)

$$\prod_{i=0}^{N-1} \Lambda_i(\text{swap}(\text{ACC}, \text{cell}[i])) \quad \Lambda_i \text{ applies swap iff } \text{tape_index}=i$$

- apply $W_{\text{r\&w}}$

$$|p\rangle_{\text{head}} \otimes |a\rangle_{\text{ACC}} \rightarrow |q\rangle_{\text{head}} \otimes |b\rangle_{\text{ACC}}$$

- swap `ACC` and `cell[i]`

tape_index

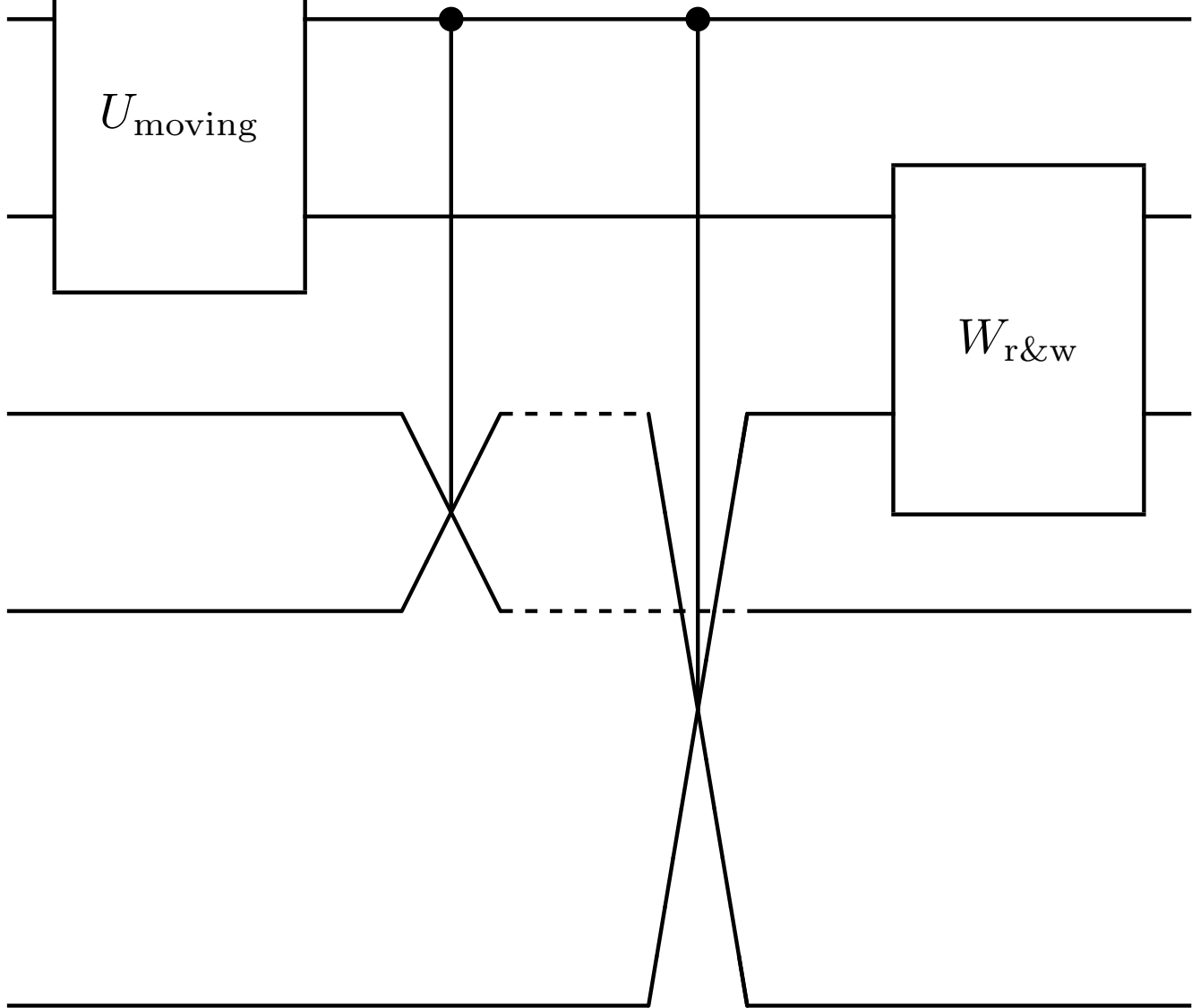
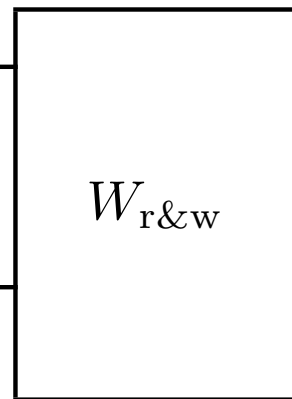
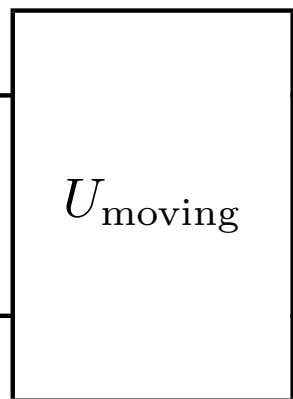
head

ACC

cell 0

⋮

cell N-1



Disadvantages of the quantum circuit

not known exactly: how many times U should be applied

desired: running time r_l depends only on input length l

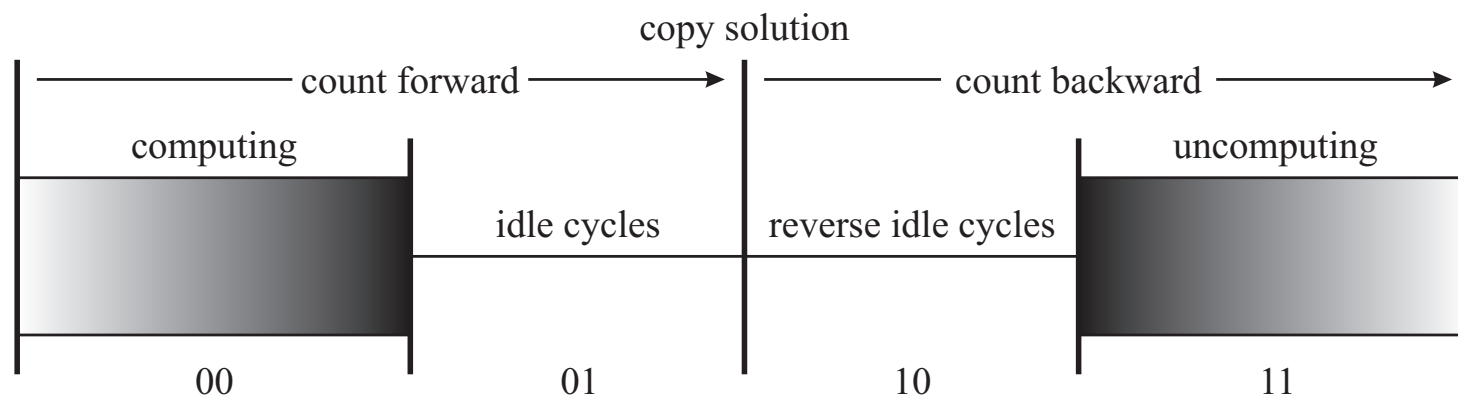
U produces garbage (e.g. does not reset ancillas)

desired:

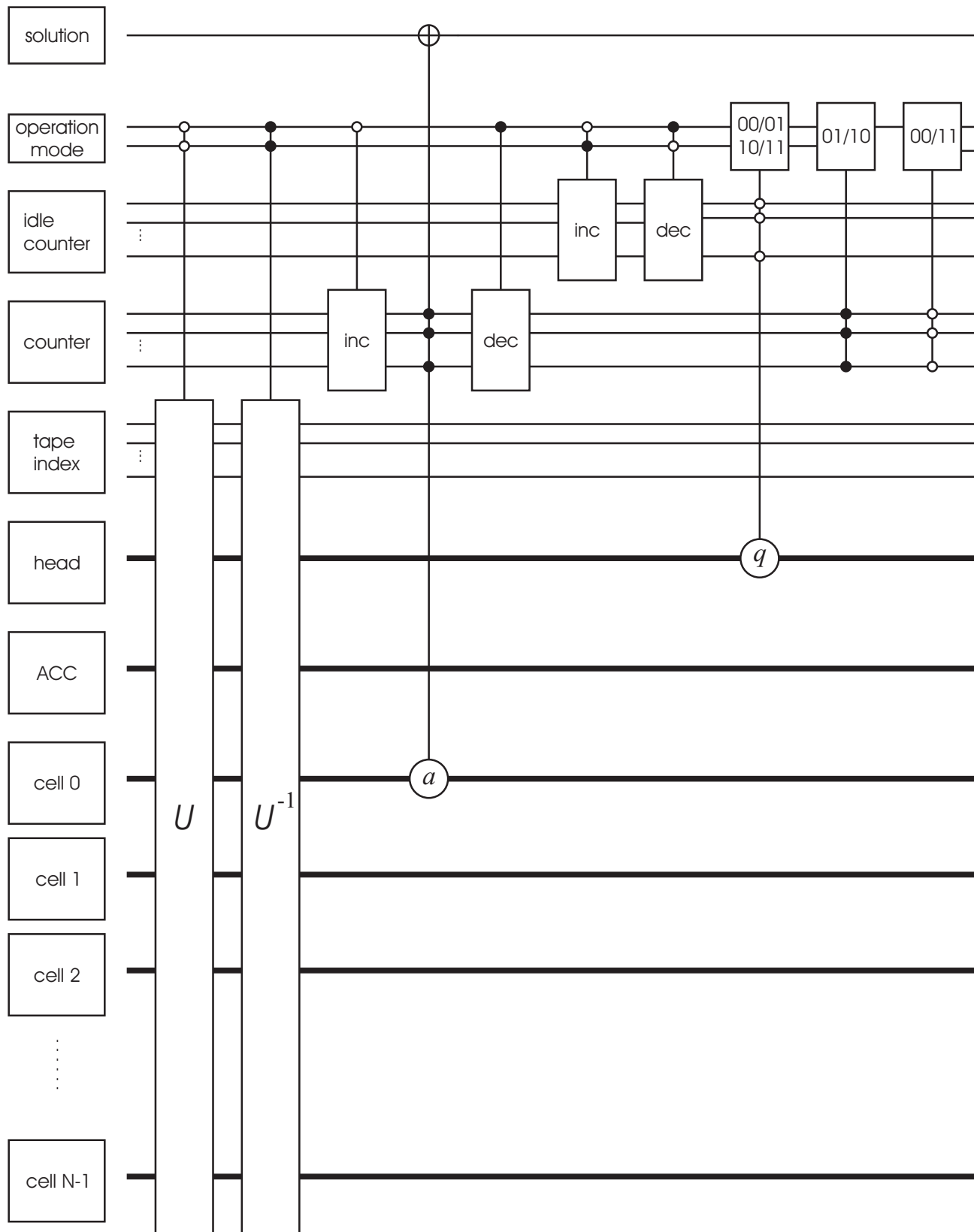
$$V^{r_l} |x\rangle \otimes |y\rangle \otimes |00\dots 0\rangle = |x\rangle \otimes |f(x) \oplus y\rangle \otimes |00\dots 0\rangle$$

$f(x)$ solution to the instance x

$\Rightarrow$ second idea: augment U and $U^\dagger$ with some control logic



Final quantum circuit



Characterization of PSPACE with quantum circuits

L arbitrary language in PSPACE



family $(V_l)_{l \in \mathbb{N}}$ of quantum circuits such that

V_l decides for all input x of length l if $x \in L$:

$$V_l^{r_l}(|x\rangle \otimes |y\rangle \otimes |00 \dots 0\rangle) = |x\rangle \otimes |y \oplus f(x) \bmod 2^{c_l}\rangle \otimes |00 \dots 0\rangle,$$

properties of V_l :

V_l permutes only computational basis states

powers of V_l produce orthogonal states (until recurrence)

$$f(x) = 0 \quad |\Psi_x\rangle, V_l|\Psi_x\rangle, \dots, V_l^{r_l-1}|\Psi_x\rangle, V_l^{r_l}|\Psi_x\rangle = |\Psi_x\rangle$$

$$f(x) = 1 \quad |\Psi_x\rangle, V_l|\Psi_x\rangle, \dots, V_l^{2^{c_l}r_l-1}|\Psi_x\rangle, V_l^{2^{c_l}r_l}|\Psi_x\rangle = |\Psi_x\rangle$$

$\implies$ orbit length depends on $x \in L$

Local Hamiltonians/observables encoding the circuits

T_k : elementary quantum gates (Toffoli gates) of V ($k = 0, \dots, s-1$)

clock: quantum register on s qubits

$$\xrightarrow{\text{green}} |1000 \dots 0\rangle \xrightarrow{\text{red}} |0100 \dots 0\rangle \xrightarrow{\text{blue}} |0010 \dots 0\rangle \mapsto \dots \mapsto |000 \dots 01\rangle \xleftarrow{\text{green}}$$

forward-time operator

$$F = T_0 \otimes |1\rangle_2 \langle 0|_1 \otimes |0\rangle_1 \langle 1|_0 + T_1 \otimes |1\rangle_3 \langle 0|_2 \otimes |0\rangle_2 \langle 1|_1 + \dots + T_{s-1} \otimes |1\rangle_1 \langle 0|_0 \otimes |0\rangle_s \langle 1|_{s-1}$$

local Hamiltonian/observable

$$A := F + F^\dagger$$

Orbit properties

F acts as a cyclic shift on $\mathcal{O} := \langle F^j |\Psi_x\rangle \mid j \in \mathbb{N} \rangle$

dimension of $\mathcal{O}$ is $d := 2^{c f(x) r s}$ depends on the solution

$\implies$ eigenvalues and eigenvectors of F restricted to $\mathcal{O}$ are

$$e^{2\pi k/d} \quad |k\rangle \quad \text{for } k = 0, \dots, d-1$$

$\implies |\Psi_x\rangle$ uniform superposition of all eigenvectors $|k\rangle$

$$|\Psi_x\rangle := \frac{1}{\sqrt{d}} \sum_{k=0}^{d-1} |k\rangle$$

F and $F^\dagger$ commute on $\mathcal{O}$

$\implies$ eigenvalues and eigenvectors of A restricted to $\mathcal{O}$ are

$$2 \cos(2\pi k/d) \quad |k\rangle \quad \text{for } k = 0, \dots, d-1$$

2 EVs with multiplicity 1 ($\equiv$ real EVs of F)

$d-2$ EVs with multiplicity 2 ($\equiv$ complex EVs of F)

Entropy of time-average

initial state:

$$|\Psi_x\rangle\langle\Psi_x| = \frac{1}{d} \sum_{k,l} |k\rangle\langle l|$$

averaging over time destroys coherence between eigenspaces

$$\bar{\rho} := \frac{1}{d} \sum_{k,l} [\cos(2\pi k/d) = \cos(2\pi l/d)] |k\rangle\langle l| = \begin{pmatrix} 1 & & & & & \\ & 1 & & & & \\ & & 1 & 1 & & \\ & & 1 & 1 & & \\ & & & & \ddots & \\ & & & & & 1 & 1 \\ & & & & & 1 & 1 \end{pmatrix}$$

$$\text{entropy } S(\bar{\rho}) = \log_2 d - (d-2)/d$$

d depends on whether x is a “yes” or “no” instance

$\implies$ difference in entropy between “yes” and “no” instance at least $c/2$ bits

c size of the output register

Measurement of A

output register consists of one qubit

prepare $|\Psi_x\rangle$

perfect measurement of A yields the results

$$2 \cos(2\pi k / (2rs)), \quad k = 0, \dots, 2rs - 1, \text{ if } x \text{ “yes” instance}$$

$$2 \cos(2\pi k / (rs)), \quad k = 0, \dots, rs - 1, \text{ if } x \text{ “no” instance}$$

with equal probability

$\implies$ we can distinguish between both cases after few samples

imperfect measurement

accuracy in the order $1/(rs)$ is sufficient to distinguish between both cases